SCOPE

INTEGRATED - CSE-IN-COLLABORATION-WITH-VIRTUSA

About the Programme

The Master of Technology (M.Tech.) in Computer Science and Engineering (CSE) is a postgraduate programme that offers advanced education in various aspects of Computer Science and Engineering. The curriculum encompasses a broad spectrum of topics including algorithms, data structures, artificial intelligence, networking, and more, often allowing students to specialize in areas like cybersecurity or cloud computing. With a focus on both theoretical knowledge and practical skills, the M.Tech CSE programme typically incorporates hands-on experiences such as laboratory sessions, projects, and internships. Additionally, students may engage in research projects, culminating in a thesis or a significant project that contributes to the field's body of knowledge. Graduates of M.Tech CSE programmes are well-prepared for diverse career opportunities in industri

Eligibility

- A candidate should have passed Higher Secondary Examination conducted by the State/ Central Board of Secondary Education or its equivalent examination with 65% aggregate in Physics, Chemistry and Mathematics (PCM).

Key Features

- 1. One of the largest departments in terms of number of faculty and students
- 2. Best-in-class infrastructure and faculty profile.

Career Prospects

- Career prospects for individuals with an M.Tech in Computer Science and Engineering (CSE) are exceptionally promising, reflecting the high demand for skilled professionals in technology-driven industries. Graduates can pursue diverse roles such as Software Development Engineers, Data Scientists, Cybersecurity Specialists, Cloud Computing Engineers, AI/ML Engineers, research scientists, IT consultants, or entrepreneurs. With expertise in advanced concepts such as Algorithms, Machine Learning, and Cybersecurity, M.Tech CSE graduates are well-equipped to excel in various sectors, including software development, finance, healthcare, and academia. Continuous learning and staying updated with emerging technologies are crucial for leveraging the myriad opportunities available and advancing in this dynamic field.

Program Outcomes

After successful completion of the program a student is expected to have abilities to

Information

PO1. Engineering knowledge:

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis:

Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions:

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems:

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage:

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society:

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability:

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8.Ethics:

Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and teamwork:

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication:

Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance:

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning:

Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Educational Objectives

- PEO1: To apply new data structures, algorithms, and process methodologies for the design and development of innovative projects of the industry and society.
- PEO2: To help comprehend knowledge of recent technologies in Data Science and Networking and use them appropriately to build frameworks leading to Full Stack Development solutions.
- PEO3: To pursue excellent leadership qualities, ethical values, and standards to become a successful entrepreneur in IT and ITES.

- PEO4: To acquire enhanced knowledge to carry out higher education in Computer Science Engineering research areas of different domains and improving their career growth path.

Programme Specific Outcome

- PSO 1: To acquire adequate knowledge on cutting edge technologies, various software tools, and techniques to solve real time problems and pursue research in related areas.
- PSO 2: To practice principles to enhance the software development processes through test driven development, behaviour driven development techniques, Cloud Services and DevOps tools for quality software.
- PSO 3: To adopt emergent computing platforms and programming languages in providing solutions to meet the industry standards.

INTEGRATED - CSE- SOFTWARE – ENGINEERING

About the Programme

MTech Software Engineering deals with giving graduates advanced conceptual understanding, technical proficiency, and the capacity to carry out independent research in the areas of project management, engineering, and software quality.The main features of this MTech Software Engineering include the focus on the application and practical uses of Software as well as development.It also deals on systematic construction of software solutions, using innovation techniques.The curriculum of MTech Software Engineering emphasises the understanding and exploitation of advanced technologies and management disciplines to support innovation.

Eligibility

Nationality

- A pass in Higher Secondary Examination conducted by the State / Central Board of Secondary Education or its equivalent examination with 65% aggregate in Physics, Chemistry and Mathematics (PCM).
- No entrance exam needed.

Key Features

1. The Software Engineering master's program is designed to equip graduates with advanced conceptual knowledge, technical skills and the ability to conduct research in the fields of software quality, engineering and project management.

2. This degree prepares the researcher for a career in an IT company, information science company and research consortia. There is a great demand for Indian software engineering graduates from all over the world.

3. SAS inks Industrial MoU with Mitsui Kinzoku Components India Private Limited (MKCI)

4. SAS inks Academic & Research MoU with CSIR-IICT

Career Prospects

The job profiles after the course MTech software engineering are Software Engineer, Senior Software Engineer, Software Developer, Senior Project Engineer, Test Engineer, Quality Assurance Engineer, etc.

Also, Average packages for MTech Software Engineering graduates are on the higher side for both private and public organizations.

The work of a Software Engineer involves initiating, planning, and completing technical and IT works.

MTech Software graduates are able to pursue a diverse array of career opportunities. There is no limit to educational programs that one can choose.

The top recruiting companies are HSBC Bank, Sandisk, Paytm, Amazon, Bloomberg, Microsoft, Google, Apple, Bain & company, etc.

Program Outcomes

After successful completion of the program a student is expected to have abilities to

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PO12. Life-long learning:

Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Educational Objectives

- PEO1: analyse, plan, design and implement high-quality software solutions for problems encountered in several domains to comply with the customer satisfaction.
- PEO2: accomplish a competitive edge in research and development and meeting out the dynamic needs of the industry and society.
- PEO3: identify and select appropriate software testing methodologies to validate software products in order to enhance its quality and reliability.
- PEO4: provide guidance in choosing adequate process models and design methodologies for the team to complete the real time projects by applying software engineering standards.

Programme Specific Outcome

- understand, analyse a problem and design an effective solution through appropriate software process models, design methodologies and software maintenance techniques.
- acquire knowledge about the new tools, software frameworks, and software documentation methods that aids in developing software projects and research.
- learn different software testing strategies, quality management and configuration techniques in order to apply for real time problems in industry and society.